Pulse - Detailed System Design (D1)

Assignment 6 - Detailed System Design, Part 1 (D1)

Part of the Assignment 4 design-diagram collection. D2 (Assignment 7) updates the capacity, retry scheduling, carrier limits, and failure handling below.

1. Header, scope, and conventions

Project title: Pulse - platform-independent IoT communication for resilient farm sensor networks.

Goal statement: Move every farm sensor reading from a field device to a trusted Hub over any available link, and keep each reading until the Hub stores it.

Team: Jacob Cohen, Sujal Choukse, Yatharth Bajaj. **Advisor:** Jeffrey Gundler. **Date:** October 6, 2026.

This document details the Node (C3) and Hub (C5); D2 details the Sensor Driver (C1), Edge (C2), Transport Adapter (C4), and Builder (C6).

The Node owns reliable delivery and sender checks. The Hub owns the data model and the storage transaction. Jacob owns C3. Yatharth owns C5. Sujal checks sensor formats with both owners. The shared Protobuf package defines I3, I5, and I6. It does not run as a component.

Figure conventions

Figure 1 uses rectangles for Hub tables. Each rectangle lists the table name and its key fields. Solid lines show foreign-key relationships. Labels give the count at both ends: `1` means exactly one, `0..*` means zero or more, and `1..*` means one or more. `PK` marks a primary key. `FK` marks a foreign key. One composite key uses all marked fields together. Lines do not show message flow. The figure uses no person or hardware symbols. The SQL source gives the full constraints. The Hub transaction enforces the minimum of one Reading per Envelope.

Basis and changes from D0

The project title, goal, component IDs, and reading key stay the same as D0 [P1]. The following changes make D0 precise.

Item	D1 decision and basis
Retry limit	Retry every 30 seconds until the 3,600-second deadline. Five retries alone do not cover US-01. This changes D0's retry limit.
Queue key	Each field Node uses `(message_id, destination)`. This prevents one destination's receipt from removing another destination's copy.
Hub model	Origin, Sensor, and Envelope tables support the existing Reading key. The Hub checks duplicate content before sending a receipt.
Carrier and identity	Current code uses Reticulum over TCP, a 16-byte identity hash, and no Pulse TTL. D0 names UDP and a 32-byte sender key.
Build status	These refinements define target behavior. Current code still uses message-only queue keys, five retries, and a flat readings table [P3].

The queue stays in memory in D1. Outage tests keep both Nodes running. Recovery after a Node restart depends on the durable queue design in D2. A local acceptance response confirms memory admission. It does not confirm storage at the Hub.

2. Data model, the D1 diagram

Figure 1 details C5. All four tables use SQLite on the Hub host.

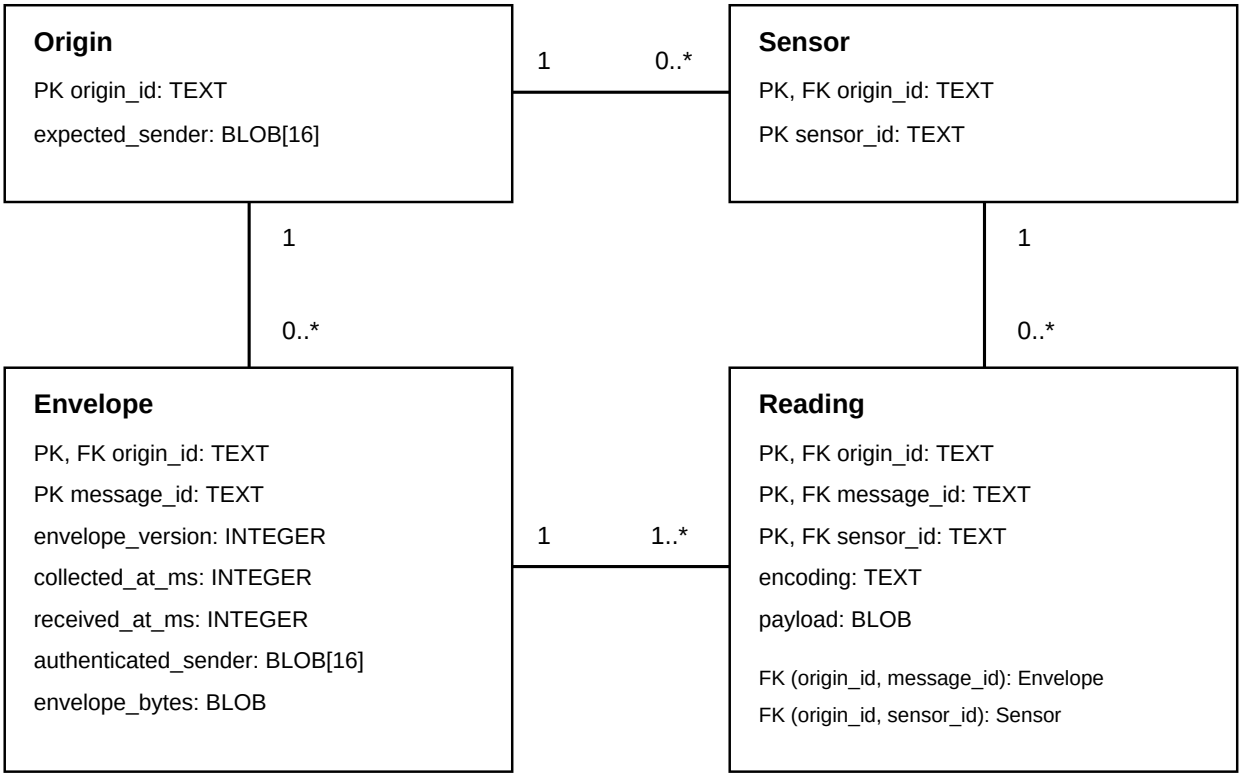
Pulse | Figure 1. Hub data model

Goal: move every farm reading to a trusted Hub and keep it until stored.

Legend: boxes = tables; lines = foreign keys; 1 / 0..* / 1..* = cardinalities.

Figure 1. C5 Hub: relational data model

Target D1 schema. PK = primary key; FK = foreign key.



A2 enforces one or more Readings per Envelope in one transaction. Lines show relationships, not message flow.

Structural decisions

Decision	Reason and requirement
Envelope is an entity	One publication groups samples into one commit and one receipt. Its stored content identifies conflicting retries. US-01, US-02; I6, I7.
Sensor is an entity	An origin can use one sensor across many messages. An encoding belongs to each Reading because formats can change. US-04.
One-to-many links	Each Envelope has one Origin. Each Reading has one Envelope and one Sensor. Reading joins their histories. US-02, US-04.
Relational store	Keys, foreign keys, and transactions protect a complete publication. A plain file needs custom checks and recovery. US-02, US-05.

An Origin represents a configured device name. Its expected sender hash binds that name to one identity. The Node trust table remains the authority. The Hub imports the same binding before it accepts remote data. One identity can own several Origin

names. Key changes require a controlled configuration update; D1 does not permit two active keys for one Origin.

Key attributes, constraints, and indexes

Table	Key attributes and rules
Origin	`origin_id: TEXT`, PK; `expected_sender: BLOB`, exactly 16 bytes. The name contains 1-64 ASCII characters. The Origin row exists before delivery.
Sensor	`(origin_id, sensor_id): TEXT`, PK; `origin_id` is an FK. Sensor rows exist before delivery. Both names contain 1-64 ASCII characters.
Envelope	`(origin_id, message_id): TEXT`, PK; `origin_id` is an FK. The stored fields are `envelope_version: INTEGER = 1`, `collected_at_ms: INTEGER >= 0`, `received_at_ms: INTEGER > 0`, `authenticated_sender: BLOB` of 16 bytes, and `envelope_bytes: BLOB`.
Reading	`(origin_id, message_id, sensor_id): TEXT`, PK. The first two fields form an FK to Envelope. `(origin_id, sensor_id)` forms an FK to Sensor. The Hub stores `encoding: TEXT` and `payload: BLOB` unchanged.

All fields are required. `collected_at_ms = 0` means the sensor time is unknown. All other times use UTC milliseconds since the Unix epoch. The receiver clock supplies `received_at_ms`. Arrival time does not replace collection time. Payload units belong to the decoder contract. For example, the moisture value is a fraction from 0 to 1.

Index	Lookup and reason
Envelope PK	Find a duplicate by origin and message in logarithmic time. US-02; I7.
Reading PK	Enforce D0's unique sample key and fetch a whole envelope. US-02, AC-01.2.
Sensor PK and Origin PK	The keys support sensor and identity checks. US-04, US-05.
Envelope `(origin_id, collected_at_ms, message_id)`	Read a device's time range without a full scan. The final field resolves equal times. US-01.
Reading `(origin_id, sensor_id, message_id)`	Fetch sensor history and check references before sensor removal. US-04.

Our planning period is 30 days. The Hub retains each Envelope and its Readings for at least 3,600 seconds after receipt. It retains duplicate keys throughout the retry period. D2 defines retention and disk monitoring. SQLite uses `foreign_keys=ON`, `journal_mode=WAL`, `synchronous=FULL`, and a 5,000 ms busy timeout. FULL adds a disk sync at each commit in WAL mode [W1].

C3 state model

`Publication` holds an immutable envelope and one unique `message_id: string` for the Node. `Delivery` holds one destination's state. One Publication has `1..*` Deliveries; each Delivery has exactly one Publication. The Delivery key is `(message_id, destination)`. Its fields are `expected_peer: bytes[16]`, `ack: AckMode`, `attempts: uint32`, `accepted_at: Instant`, `deadline: Instant`, `next_attempt: Instant`, `wrapper: bytes`, and `notify: set<uint64>`. Retries use a monotonic clock so a system-clock adjustment does not move their deadline.

The Node uses a hash map for Deliveries and one queue for new work. Capacity includes queued and waiting reliable Deliveries. The field limit is 128 Deliveries. Best-effort traffic uses a separate 64-item queue. A full reliable queue rejects new work. A full best-effort queue removes its oldest item and increases `dropped`. Durable queue storage remains part of D2.

3. Core algorithms

A1. Retry delivery until the requested receipt or expiry

1. Purpose. A1 keeps reliable publications during link outages. It prevents an early receipt from deleting data before the Hub commit. It implements US-01 and US-02 through I3, I4, I5, and I6.

2. Inputs and outputs. Inputs are `PublishRequest`, verified `StatusUpdate`, `now: Instant`, and configured limits. Section 5 gives exact wire types. Output is `Result<bool, ErrorDetail>`, zero or more signed packet sends, and `StatusEvent` values. Status values are `RECEIVED`, `STORED`, `EXPIRED`, or `FAILED`.

3. Procedure and complexity. The Node checks the request and encodes its signed wrapper. It rejects remote wrappers above 383 bytes. It rejects duplicate message IDs with different envelope bytes. It admits one Delivery per destination. A repeat of the same pending request reuses the Delivery without extending its deadline. The Node adds the caller to its notification set. The first attempt occurs immediately. The deadline is acceptance plus 3,600 seconds.

At each one-second tick, the Node checks expiry before sending. It expires Deliveries whose deadline is at or before now. It sends each remaining due wrapper once. It increases the attempt count and sets the next attempt to now plus 30 seconds. A send error keeps the Delivery. A delayed tick sends once; it does not send a burst of missed attempts. The interval is a minimum; the carrier can delay a send.

The Node checks each remote receipt's signature and expected peer. It resolves the destination from that peer's configured service. For `ACK_STORED`, `RECEIVED` does not remove the Delivery. A valid `STORED` removes only that destination's Delivery. For `ACK_RECEIVED`, a valid `RECEIVED` or `STORED` completes it. A remote `FAILED` or `EXPIRED` does not prove storage. The Node keeps an `ACK_STORED` Delivery until local expiry. It deletes a Publication only when it has no Deliveries.

Admission and receipt lookup take expected `O(1)` plus `O(B)` encoding and signature work. A tick takes `O(Q + D*B)`, where Q is pending Deliveries, D is due sends, and B is wrapper bytes. Memory takes `O(Q*B)` plus notification sets. The scan uses library containers; it does not require custom sorting.

Planning size	Estimated work and effect
One field Node	One message per 60 seconds gives at most 60 pending messages in one hour, with one destination. A tick checks about 60 entries.
100 times that load	A shared test Node holds 6,000 Deliveries. A tick checks 6,000 entries. Its test capacity is 8,192. This does not fit the field limit.
Storage estimate	At the 383-byte wrapper limit, 60 wrappers use about 23 KB; 6,000 use about 2.3 MB, before container overhead.

4. Why this algorithm. A simple scan suits the bounded field queue. A minimum heap reduces idle checks to `O(log Q)` updates but adds cancellation state. Stop-and-wait blocks other sensors during an outage. Five retries end after about 150 seconds and cannot cover US-01. A heap becomes an option if measured scheduler cost limits performance at 100 times load. Radio airtime can dominate these CPU costs.

5. Edge cases. An empty queue sends nothing. Equal retry times each get one send. Their order does not affect storage. Repeated receipts after completion do nothing. A wrong peer, bad signature, or unknown message does not remove data. Full capacity returns `QUEUE_FULL` without removal. At exactly the deadline, expiry wins. With several services on one peer, each destination uses a distinct message ID because v1 receipts omit destination. Node restart loses memory state; persistent storage remains part of D2.

A2. Check and commit a complete envelope once

1. Purpose. A2 converts a verified delivery into one stored row per sensor. It sends `STORED` only after a complete commit. It implements US-02, US-04, and US-05; AC-01.1 to AC-01.3; I6 and I7.

2. Inputs and outputs. Input is `DeliveryEvent{destination: string, message: PulseEnvelope, authenticated_sender: bytes[16]}` plus `receivedAtMs: int64 > 0`. Output is `StoreResult = STORED | DUPLICATE` or

`StoreError`. `StoreError` is `INVALID_ENVELOPE | UNKNOWN_SENSOR | UNSUPPORTED_ENCODING | INVALID_PAYLOAD | IDENTITY_MISMATCH | CONFLICTING_DUPLICATE | STORAGE_ERROR`. These are internal codes, not new Protobuf enum values. Success causes `AckRequest{message_id, status: STORED, detail: ""}`. Failure causes a log entry and no receipt.

3. Procedure and complexity. The Hub checks the envelope fields and sender binding. It checks each sensor against the Sensor table. It rejects repeated sensor IDs. The decoder registry selects a library parser for each encoding. If any sample fails validation, the Hub rejects the whole envelope.

The Hub encodes the parsed envelope with the pinned Java Protobuf codec. It stores this representation as `envelope_bytes`. This uses the same codec for every comparison; it does not compare arbitrary Protobuf field order on the network. The Hub starts one `BEGIN IMMEDIATE` transaction. It looks up `(origin_id, message_id)`. If the key exists, it compares stored bytes and sender with the new values. A match returns `DUPLICATE`. A difference causes rollback and returns `CONFLICTING_DUPLICATE`. A retry must keep sample order and payload bytes unchanged.

For a new key, the Hub inserts one Envelope and all its Readings. It commits the transaction, returns `STORED`, and sends the receipt. A failed commit rolls back all new rows. A failed receipt send leaves committed rows unchanged. The next matching retry returns `DUPLICATE` and sends the receipt again. The PK check inside the transaction handles concurrent writers. The transaction reports data conflicts rather than ignoring them.

Let S be samples, B be envelope bytes, R be stored Readings, and E be stored Envelopes. Checks take `O(B + S)` with a set for sensor IDs. Indexed inserts take `O(S log R + log E)`. Memory takes `O(B + S)`. Commit latency also depends on disk sync.

Planning size	Estimated work and effect
Initial farm	10 devices, one sample per minute: 14,400 Readings/day and 432,000 in 30 days. Average intake is 0.167 envelopes/second.
100 times that load	1,000 devices: 1,440,000 Readings/day and 43,200,000 in 30 days. Average intake is 16.67 envelopes/second.
Index and disk estimate	`log2(R)` rises from about 19 to 26. A planning allowance of 1 KB per sample gives about 432 MB or 43.2 GB.

The estimates use 10 devices with one sensor each. Allow extra disk space for indexes and WAL. The larger case needs a storage and commit benchmark; radio capacity is a separate limit.

4. Why this algorithm. A transaction prevents partial envelopes. A primary key and full content comparison give persistent duplicate checks. An in-memory set loses history after restart. A separate lookup followed by an unlocked insert races with another writer. A receipt before commit loses data after a crash. Generated Protobuf bytes avoid a custom hash or canonical JSON scheme.

5. Edge cases. The Hub rejects empty samples, unknown encodings, empty sensor IDs, negative times, and repeated sensor IDs. Equal times remain separate messages. Queries order results by time, then message ID. Unknown collection time stays zero. An identical retry returns `DUPLICATE`. Changed payload under the same key returns a conflict. Unapproved remote identities never reach storage. Disk full, busy timeout, and a failed commit send no receipt. A crash before commit leaves no new envelope; a crash after commit permits a matching retry.

4. Build-versus-reuse decisions

These choices use the D1 dependency snapshot [P3]. Rust versions are pinned in Cargo.lock and Java versions in gradle/libs.versions.toml. We build the delivery rules and reuse libraries for codecs, storage, and cryptography.

Piece	Build or reuse; license	Maturity, performance, and fit
Delivery lifecycle	Build; Pulse course license	Project-specific retry and receipt rules. Bounded scans suit 60 pending messages. Library containers provide the basic operations.

Piece	Build or reuse; license	Maturity, performance, and fit
Atomic storage and duplicate rules	Build; Pulse course license	The key, transaction order, and content conflict rule belong to Pulse. SQLite provides transaction recovery.
Decoder registry and sensor checks	Build; Pulse course license	US-04 needs a small encoding-to-decoder map. Each decoder defines its fields and units.
Local RPC and framing	Build the contract; Pulse course license	I3/I6 need a shared small frame, explicit errors, and Rust/Java vectors. Generated codecs encode the body.
Async I/O and timers	Reuse Tokio 1.53.1; MIT [W2]	Maintained runtime with documented support policy. Bounded channels and timers fit OS hosts. ESP32 runtime fit remains a D2 task.
Protobuf codecs	Reuse prost 0.13.5, Apache-2.0; protobuf-java 4.32.1, BSD-3-Clause [W3, W4]	Established schema tools produce compact binary messages. Shared vectors check language agreement. Generated bindings preserve stable field numbers.
Signature and verification	Reuse ed25519-dalek 3.0.0; BSD-3-Clause [W5]	Established library family; version 3 matches the SDK. The library signs exact bytes. D2 covers board measurements.
Network stack	Reuse reticulum-sdk 2.3.4; MIT [W6]	Small upstream project; direct LoRa support is experimental. Our dependency pins this version. It provides paths and encryption. Hardware and Python interoperation remain required checks.
Database engine and driver	Reuse SQLite, public domain; Xerial 3.49.1.0, Apache-2.0 and BSD-2-Clause notices [W7, W8]	Mature embedded database; maintained JDBC driver. One writer fits initial intake. Host disk speed determines FULL commit latency.
JSON decoding	Reuse Gson 2.13.1; Apache-2.0 [W10]	Maintenance mode is sufficient for JSON parsing. The decoder uses strict parsing and explicit field checks. Adding it to runtime is a D1 target; it is a test dependency today.

We retain dependency license notices. The smaller Reticulum SDK needs particular attention to version pinning and interoperability tests.

5. API contract

Transport, session, and common types

I3 and I6 use persistent TCP on `127.0.0.1:4401` by default. Configuration can select another loopback port from 1 to 65,535. One frame contains a 4-byte unsigned big-endian body length and one Protobuf `RpcEnvelope`. Body length is 1 to 1,048,576 bytes. The Node checks the length before allocating the body. A bad length closes the connection. These methods use protocol error codes, not HTTP status codes.

The common wrapper is `RpcEnvelope{protocol_version: uint32 = 1, request_id: uint64, op: Operation, body: oneof}`. A request uses a required ID from 1 to 2^64-1, unique among that connection's pending requests. A response echoes that ID. An event uses ID zero. `op` and `body` must agree. Proto3 defaults do not make an application-required value optional.

Type or field	Exact rule
`Name`	Required ASCII string, 1-64 characters, using letters, digits, _, or -. Applies to origin, sensor, service, and destination names. Service `node` is reserved. D1 uses service names, not arbitrary hashes.
`message_id`	Required canonical lowercase UUID v4 string, 36 characters. Retries keep it unchanged. Within one Node it identifies one immutable Publication.

Type or field	Exact rule
`PulseEnvelope`	`{envelope_version:uint32, message_id:string, origin_id:string, collected_at_ms:int64, samples:list<SensorSample>}`. Version is 1. Origin is Name. Time is optional, 0 when unset; otherwise 1 to 2^63-1 UTC milliseconds.
`SensorSample`	`{sensor_id:string, encoding:string, payload:bytes}`. Sensor is Name and is unique within the envelope. Encoding is required, 1-128 ASCII characters, and names an installed decoder.
Samples and payload	Required list with 1-16 samples. Each payload contains 1-262,144 bytes. The total local frame must fit the frame limit. Remote publication also requires its complete signed wrapper to fit 383 bytes.
`AckMode`	`ACK_NONE=0`, `ACK_RECEIVED=1`, `ACK_STORED=2`. Optional; default is `ACK_NONE`. Other values are invalid. Farm telemetry explicitly requests `ACK_STORED`.
`Response`	Success: `{ok:true, error:absent, health:absent}`; HEALTH adds HealthStatus. Failure: `{ok:false, error:{code:ErrorCode,message:string}, health:absent}`. The message explains the cause; callers use the code.

I5 carries `TransportPacket{payload:bytes, sender_identity:bytes[16], signature:bytes[64]}`. The payload contains `SignedPayload{ack:AckMode, body:data(bytes) | status(StatusUpdate)}`. Data bytes encode one PulseEnvelope. The signature covers the exact payload bytes. The configured public identity contains 64 bytes, shown as 128 hex characters. The 16-byte hash refers to that identity; it is not the full public key. Reticulum owns the route and hop limit [P3].

The 383-byte cap comes from the pinned SDK's encrypted single-packet limit. It includes the whole Pulse wrapper. A 1 MiB local frame does not imply a 1 MiB radio packet. Oversize remote data returns `INVALID_MESSAGE`; D2 can define fragmentation if required.

Key operations on C3

Method	Inputs, required values, and limits	Output and errors
`PUBLISH` / I3	Required `destination:Name`, `message:PulseEnvelope`. Optional `ack:AckMode=ACK_NONE`, `want_acceptance:bool=false`. Sender origin must match the local configured origin.	Success returns Response if acceptance is true. Failure always returns Response. `INVALID_MESSAGE`: bad fields, conflicting pending ID, unsupported enum, or oversize wrapper. `SERVICE_UNAVAILABLE`: no configured provider. `QUEUE_FULL`: reliable capacity is full. `UNAUTHORIZED`: origin binding fails.
`SUBSCRIBE` / I6	Required `service:Name`; it must name a service allowed for this local component. Repeat registration on the same connection has no extra effect. Only one active Hub consumer owns a storage service.	Success returns Response. Deliveries then use Event. `INVALID_MESSAGE`: invalid or reserved service. `SERVICE_UNAVAILABLE`: another consumer owns it. `UNAUTHORIZED`: the component cannot host it.
`ACK` / I6	Required `message_id:string` under the UUID rule; required `status:DeliveryStatus=STORED`. Optional `detail:string=""`, 0-256 UTF-8 bytes. The calling connection must own the service's pending storage delivery.	Success returns Response and causes a signed remote storage receipt. `INVALID_MESSAGE`: unknown waiter, non-STORED status, or bad fields. `UNAUTHORIZED`: another connection attempts the receipt. A lost remote receipt triggers a new delivery; its new waiter permits another ACK.
`HEALTH` / I3 or I6	Required empty `HealthRequest{}`; no parameters. Wrapper version and request ID rules still apply.	Success returns Response with HealthStatus. `INVALID_MESSAGE`: version or operation/body mismatch. A closed connection yields a client I/O error.

The implementation snapshot [P3] still needs the name limits, origin binding, and ACK ownership checks described here. Loopback restricts network access; it does not authenticate every local process. The first deployment uses one trusted service account and protected configuration files.

Key method inside C5

`ReadingStore.store(envelope:PulseEnvelope, authenticatedSender:bytes[16], receivedAtMs:int64)` returns `StoreResult = STORED | DUPLICATE`. All arguments are required. The limits match the common types. `receivedAtMs` is 1 to 2^63-1 UTC milliseconds. A2 defines the checks and transaction order. The method generates comparison bytes from the parsed envelope with the pinned Java Protobuf codec.

Internal error	Cause and caller action
`INVALID_ENVELOPE`	A field, count, time, or duplicate sensor fails a rule. The Hub logs the error and sends no receipt.
`UNKNOWN_SENSOR` / `UNSUPPORTED_ENCODING` / `INVALID_PAYLOAD`	A sensor is not configured, a decoder is absent, or its parser/check fails. The Hub rejects the whole envelope.
`IDENTITY_MISMATCH` / `CONFLICTING_DUPLICATE`	The identity binding fails, or the same key carries different bytes. The Hub keeps committed data and sends no receipt.
`STORAGE_ERROR`	SQLite returns busy timeout, disk full, or another transaction error. The Hub rolls back and sends no receipt. The origin retries until expiry.

The current Java method returns Outcome and throws SQLException. The D1 method adds the listed checks and internal errors. These errors stay inside C5; they do not add unassigned wire enum values.

Events, status binding, and error recovery

Output	Exact shape, units, and conditions
`DeliveryEvent` / I6	`{destination:string, message:PulseEnvelope, authenticated_sender:bytes}`. Destination is the local service hash. Sender is exactly 16 bytes for remote data or empty for local data. C5 accepts an empty sender only for a configured local origin.
`StatusUpdate` / I5	`{message_id:string, status:DeliveryStatus, detail:string}`. Detail has at most 256 UTF-8 bytes. A signed expected peer supplies it. Completion requires a matching peer and pending service.
`StatusEvent` / I3	`{message_id:string, status:DeliveryStatus, detail:string}`. `RECEIVED=1`, `STORED=2`, `EXPIRED=3`, `FAILED=4`. Zero is invalid.
`HealthStatus`	`{inventory_id:string, boot_id:string, uptime_seconds:uint64, component:string, queue_depth:uint32, dropped:uint64, last_contact_ms:int64}`. Inventory is Name; boot ID is UUID v4; component is `node`. Queue depth counts pending Deliveries, including waiting items. Dropped counts best-effort removals since boot. Contact is epoch milliseconds, zero before any verified peer contact.

For local deliveries, C5 resolves an empty sender to the configured local identity before storage. The schema always stores a 16-byte identity hash. Collection and receipt times remain distinct. Status events omit destination in v1. Therefore, the Edge assigns distinct IDs to different destinations when one peer provides several services. A later schema can add explicit destination binding.

Error code or failure	When it occurs	Caller response
`INVALID_MESSAGE=1`	A supported frame has invalid Protobuf, version, operation/body, or application fields.	The caller corrects the request before retrying. A malformed body gets request ID zero if no ID can be read.
`SERVICE_UNAVAILABLE=2`	A destination lacks a provider or the storage service has another active owner.	The caller retries when the configured service becomes available.
`QUEUE_FULL=3`	Pending reliable Deliveries reach capacity.	The caller retains the sample and retries admission later. The Node does not own a rejected sample.
`UNAUTHORIZED=4`	A readable local request fails an origin, service, or ACK ownership check.	The caller corrects its configuration or identity before retrying.
Invalid remote sender	Peer is unknown, signature is bad, or origin/service binding fails.	Receiver drops and logs the packet. It sends no trusted negative receipt. This implements AC-01.3 without replying to an untrusted sender.
Invalid length or closed stream	Zero/oversize frame, truncated stream, or I/O failure.	The client reconnects after 100 ms and doubles the delay to a 5,000 ms cap. It subscribes again. No event replay exists on the connection.

A valid `STORED` is the only success that proves Hub storage for `ACK_STORED`. Expiry means storage is unconfirmed; it does not prove that the Hub has no copy. A missing receipt after commit can produce this result. The Node reports the result so the technician can investigate.

Versioning. `protocol_version` versions local RPC; `envelope_version` versions data; both start at 1. Compatible changes add optional Protobuf fields and preserve existing field numbers. Changing a field type, required rule, receipt meaning, or supported identity width breaks the contract. Such a change requires a version increase and coordinated peer upgrade. D0's identity-width change needs no migration of deployed farm devices because D0 is a design baseline. Future deployed v1 changes require a migration. A new sensor with an installed decoder does not require a new envelope version.

Example request and response

The example shows Protobuf JSON notation for readability. The wire uses Protobuf bytes with the length prefix. Bytes use base64. JSON uses strings for 64-bit integer fields. The sample payload decodes to `{"moisture":0.42}`. Moisture uses a fraction, not a percentage.

```
{
  "protocolVersion": 1,
  "requestId": "17",
  "op": "PUBLISH",
  "publish": {
    "destination": "hub-main",
    "message": {
      "envelopeVersion": 1,
      "messageId": "11111111-2222-4333-8444-555555555555",
      "originId": "field-01",
      "collectedAtMs": "1788912000000",
      "samples": [{
        "sensorId": "soil-1",
        "encoding": "application/json",
        "payload": "eyJtb2lzdHVyZSI6MC40Mn0="
      }]
    },
    "ack": "ACK_STORED",
    "wantAcceptance": true
  }
}

{
  "protocolVersion": 1,
  "requestId": "17",
  "op": "RESPONSE",
  "response": { "ok": true }
}
```

After the Hub commits, the field Node sends this event to the Edge.

```
{
  "protocolVersion": 1,
  "requestId": "0",
  "op": "EVENT",
  "event": { "status": {
    "messageId": "11111111-2222-4333-8444-555555555555",
    "status": "STORED",
    "detail": ""
  } }
}
```

An admission failure instead returns `response:{ok:false,error:{code:QUEUE_FULL,message:"reliable outbound queue full"}}` under request ID 17. Acceptance and `STORED` remain separate results.

6. Technology choices with justification

Database: SQLite through Xerial JDBC

We chose SQLite because the Hub needs one local writer, transactions, and duplicate checks without another service to install. WAL supports concurrent reads. Xerial provides the Java driver [W1, W7, W8]. PostgreSQL would add setup and administration that the initial workload does not need.

Backend: Rust Node with Tokio; Java 21 Hub

Rust and Tokio provide bounded queues, timers, and asynchronous I/O for the Node. Java 21 gives the Hub a straightforward JDBC storage path. The JVM runs only on the Hub host. We kept these existing project choices rather than adding a web framework or rewriting the Node in another language.

Interface: command-line tools and logs

Technicians configure devices and inspect delivery faults through commands, logs, and HEALTH. The user stories do not require a dashboard, so we are not building one.

Messaging and hardware: Protobuf, Reticulum, and ESP32-S3

Protobuf defines the shared Rust/Java messages. Reticulum provides the carrier layer, starting over TCP before board testing. We use the existing ESP32-S3 and LoRa hardware. We chose this path over a central MQTT broker because field delivery must work without a separate IP service. D2 defines the LoRa packet limits and board recovery design.

Hosting: local Linux host with systemd

The prototype runs on an existing Linux host. systemd restarts the Node, Edge, and Hub independently. A local database keeps farm internet access out of the collection path. We use the $200 course budget and existing equipment [P4]; any additional purchase must fit that cap.

Project sources

- **P1:** `documents/sd1/Design_Diagrams/D0_High_Level_Design.md`, with its original figure sources. D0 provides the component and interface baseline.
- **P2:** `User_Stories.md`: US-01 to US-05, UC-01, and AC-01.1 to AC-01.3.
- **P3:** The sibling `monorepo`, commit `c845c4e`, including: `node/src/buffer.rs`, `node/src/node.rs`, `node/src/rpc.rs`, `hub/.../ReadingStore.java`, `HubMain.java`, `DecoderRegistry.java`, and `common/protocol/proto/pulse/v1/*.proto`. Dependency sources: `Cargo.toml`, `Cargo.lock`, `gradle/libs.versions.toml`, `MODULE.bazel`, and `LICENSE`.
- **P4:** `documents/sd1/assignments/assignment-3/Constraints_Essay.md`: $200 course budget and existing board inventory.
- **P5:** `docs/meetings/2026-09-11-first-advisor-meeting.md`: desktop-first plan and board tests.

External sources and license checks

Dependency documentation and license sources follow.

ID	Source and supported decision
W1	SQLite WAL: commit sync and single-writer behavior.
W2	Tokio: runtime support and MIT license.
W3	prost: generated Rust codecs and Apache-2.0 license.
W4	Protobuf: schema tools; license.
W5	ed25519-dalek: signature implementation. Pinned 3.0.0 manifest states BSD-3-Clause.
W6	Reticulum SDK: MIT terms and experimental direct LoRa support. Pinned 2.3.4 packet source defines the 383-byte limit.
W7	SQLite copyright: public-domain terms.
W8	Xerial JDBC: maintenance, supported hosts, and license files.
W10	Gson: maintenance mode and Apache-2.0 license.